

Physics Lab Summary

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1 Propagation of Uncertainty

We use the *Quadrature Method* to propagate uncertainty. That is, if a quantity f is made of two measured quantities x and y , the uncertainty of f is given by $\delta f = \sqrt{(\frac{\partial f}{\partial x} \delta x)^2 + (\frac{\partial f}{\partial y} \delta y)^2}$.

2 How Many Significant Figures to Keep?

You shouldn't keep more significant figures than the uncertainty of your measurement. For example, if δx has 2 sig figs and δy has 1 sig fig, then your f should only have 1 sig fig.

3 The Uncertainty of the Measured Slope

How do we calculate the uncertainty of the measured slope?

We first want to calculate the uncertainties in the residuals:

Uncertainty in Residuals:

$$\sigma = \sqrt{\frac{1}{N-p} \sum_{i=1}^N r_i^2}$$

After calculating σ , we can then calculate the uncertainty in the fitted slope. Interestingly, the equations are different for $y = mx$ and $y = mx + b$.

Uncertainty in Fitted Slope (this is the formula for $y=mx+b$)

$$\sigma_m = \frac{\sigma}{\sqrt{\sum_{i=1}^N (x_i - \bar{x})^2}}$$

Uncertainty in Fitted Slope (this is the formula for $y=mx$)

$$\sigma_m = \frac{\sigma}{\sqrt{\sum_{i=1}^N (x_i)^2}}$$

However, we don't need to calculate this manually. The covariant matrix in Scipy already calculate it automatically for us. How that works is a very interesting topic that I should write about at another time.

4 Analysis of Residuals

A good residual graph should have the residuals spread evenly above and below the zero line. If you see a pattern in the residuals that is not random (i.e., a quadratic, cubic, or sinusoidal pattern), that probably means that your model didn't capture some of the physics.

Sometimes, this occurs because you didn't include an offset in your model (just like what happened in experiment 11-12). A good way to guess whether this pattern is caused by the offset is to add this offset into the model and do a Taylor expansion and compare the resulting equation to your original equation (see experiment 11-12 for more information).

5 Dealing with Imperfections in Instrumentation

Sometimes the instruments we use are not perfect. For example, the current provided by the power supply might fluctuate even in constant current mode (especially when the current is small). When we encounter this type of scenario we want to take multiple data points around a single current value (let's say 5 data points around 0.5 A), and then take the average and standard deviation of these 5 data points. We use the average as our x data point and the standard deviation of these 5 data points as the uncertainty for this x data point. This is a more systematic way than just taking one data point at a specific current value then eyeballing the fluctuations of the power supply to use the fluctuations as uncertainty.

6 Does Our Data Make Sense?

Suppose we have 2 methods to measure how one quantity (let's say the angular displacement of the torsional oscillator) varies with another quantity. One method (let's say using LoggerPro to measure change in voltage) provides more accuracy but takes more time and is more complicated. The other method (let's say eyeballing) is faster but less accurate. Which one should you use?

My preference is to use the quick method first to get a sense of what the relationship should be. Then, if you have time left, use the slow, complicated

method to get more accurate data. This is because you might mess up some steps in the complicated method and end up get a wrong relationship. However, if you have the results from the quick method to serve as a benchmark, you can tell if you messed up or not.

Also, check our data points once in a while when you are taking them. For example, let's say we are still measuring how angular displacement varies with current. Does the angular displacement look symmetrical when we reverse the current direction? If not, we might have messed up the instruments somehow (like how we wrapped our string in one direction is different from how we wrapped our string in the other direction). When we see an anomaly, stop and check our instrumentation. We want to discover mistakes early on.

7 We Should Cheer when Something Goes Wrong

When we are not getting what we expect to get, we should cheer! This is often the best time to learn because we are either making some mistake that nobody else is making, or discovering something that nobody else discovered. These are very good opportunities to build intuition!

8 Don't let go of Things that Bother You

Trust our intuition. When something bothers us (that doesn't seem to bother other people), it probably means that we are onto something. Just dig a little bit deeper (take a few guesses, look it up on the internet, talk to people) and we will probably find something surprising. This happened for me when I was bothered by the butterfly-shaped curve of how current varied with voltage (in lab 3) and when I was bothered by how the residuals for lab 11-12 had a quadratic shape. It is very satisfying when I figured out why.

9 Keep Good Documentation & Develop Your Toolkit

Document our questions, thoughts & observations regularly and in detail. Reflect on them regularly as well. They are valuable sources of intuition.

Develop and refine our toolkit regularly. I can't stress how much organizing my python toolkit has helped me when setting up jupyter notebooks for data analysis. Think about ways to systematize and optimize our toolkit as well.

Note: when our code runs into an error and we don't know what is happening (even asking AI doesn't help), search up and read the documentation of the code. The seemingly slowest way to debug often turns out to be the fastest. After we figured out what's happening, document this in our toolkit so when

we run into a similar problem next time, we know what's going on & how to fix it.